Lakshya JEE 2.0 (2024)

Limit, Continuity and Differentiability

DPP-03

1. $\lim\limits_{x\to\infty}\dfrac{x^2+x+1}{3x^2+2x-5}$ is equal to:

(1) $\frac{1}{3}$ (2) $\frac{2}{3}$
(3) $\frac{4}{3}$ (4) $\frac{5}{3}$

2. $\lim\limits_{x\to\infty}\left(\sqrt{x^2+x+1}-\sqrt{x^2+1}\right)$ is equal to:

(1) 2 (2) 1
(3) $\frac{1}{2}$ (4) $\frac{1}{4}$

3. $\lim\limits_{x\to\infty}\dfrac{2x+7\sin x}{4x+3\cos x}=$

(1) –1 (2) 1
(3) $\frac{1}{2}$ (4) $-\frac{1}{2}$

4. $\lim\limits_{x\to\infty}\dfrac{(2x+1)^{40}(4x-1)^5}{(2x+3)^{45}}=$

(1) 16 (2) 24
(3) 32 (4) 8

5. The value of limit $\lim\limits_{x\to\infty}\left(x-\sqrt{x^2+x}\right)$ is:

(1) –1/2 (2) 1/2
(3) 1 (4) –1

6. $\lim\limits_{x\to-\infty}\dfrac{x^2\sin\left(\frac{1}{x}\right)}{\sqrt{9x^2+x+1}}$ is equal to:

(1) $\frac{1}{3}$ (2) $-\frac{1}{3}$
(3) $\frac{-2}{3}$ (4) $\frac{2}{3}$

7. $\lim\limits_{x\to\infty}\left(\dfrac{x^3}{3x^2-4}-\dfrac{x^2}{3x+2}\right)$ is equal to

(1) does not exist (2) 1/3
(3) 0 (4) 2/9

8. $\lim\limits_{x\to0}\dfrac{(27+x)^{\frac{1}{3}}-3}{9-(27+x)^{\frac{2}{3}}}$ equals

(1) $\frac{1}{3}$ (2) $-\frac{1}{3}$
(3) $-\frac{1}{6}$ (4) $\frac{1}{6}$

9. If $f(4)=4$, $f'(4)=1$, then $\lim\limits_{x\to4}\dfrac{2-\sqrt{f(x)}}{2-\sqrt{x}}$ is equal to:

(1) 0 (2) 1
(3) –1 (4) 2

10. $\lim\limits_{x\to\infty}\left(\dfrac{x^2+2x-1}{2x^2-3x-2}\right)^{\frac{2x+1}{2x-1}}$ is equal to

(1) 0 (2) ∞
(3) 1/2 (4) None of these

11. The value of $\lim\limits_{n\to\infty}\dfrac{1^3+2^3+3^3+......+n^3}{\left(n^2+1\right)^2}$ is:

(1) $\frac{1}{4}$ (2) $\frac{1}{2}$
(3) $\frac{1}{2\sqrt{2}}$ (4) none of these

12. $\lim\limits_{x\to0}\dfrac{e^{\tan x}-e^x}{\tan x-x}$ is equal to:

(1) 2 (2) 1
(3) 3 (4) 4

13. $\lim\limits_{n\to\infty}\left[\dfrac{1}{1-n^2}+\dfrac{2}{1-n^2}+\dfrac{3}{1-n^2}+.........+\dfrac{n}{1-n^2}\right]$ is equal to

(1) 0 (2) $-\frac{1}{2}$
(3) $\frac{1}{2}$ (4) None of these

14. If $f: R \to R$ is given by $f(x) = x+1$, then the value of

$$\lim_{n\to\infty} \frac{1}{n}\left[f(0) + f\left(\frac{5}{n}\right) + f\left(\frac{10}{n}\right) + \ldots + f\left(\frac{5(n-1)}{n}\right)\right]$$

is:

(1) $\frac{3}{2}$ (2) $\frac{5}{2}$

(3) $\frac{1}{2}$ (4) $\frac{7}{2}$

15. $\lim_{\alpha \to \pi/4} \dfrac{\sin\alpha - \cos\alpha}{\alpha - \frac{\pi}{4}} =$

(1) $\sqrt{2}$ (2) $1/\sqrt{2}$

(3) 1 (4) None of these

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. **(1)**
2. **(3)**
3. **(3)**
4. **(3)**
5. **(1)**
6. **(2)**
7. **(4)**
8. **(3)**
9. **(2)**
10. **(3)**
11. **(1)**
12. **(2)**
13. **(2)**
14. **(4)**
15. **(1)**